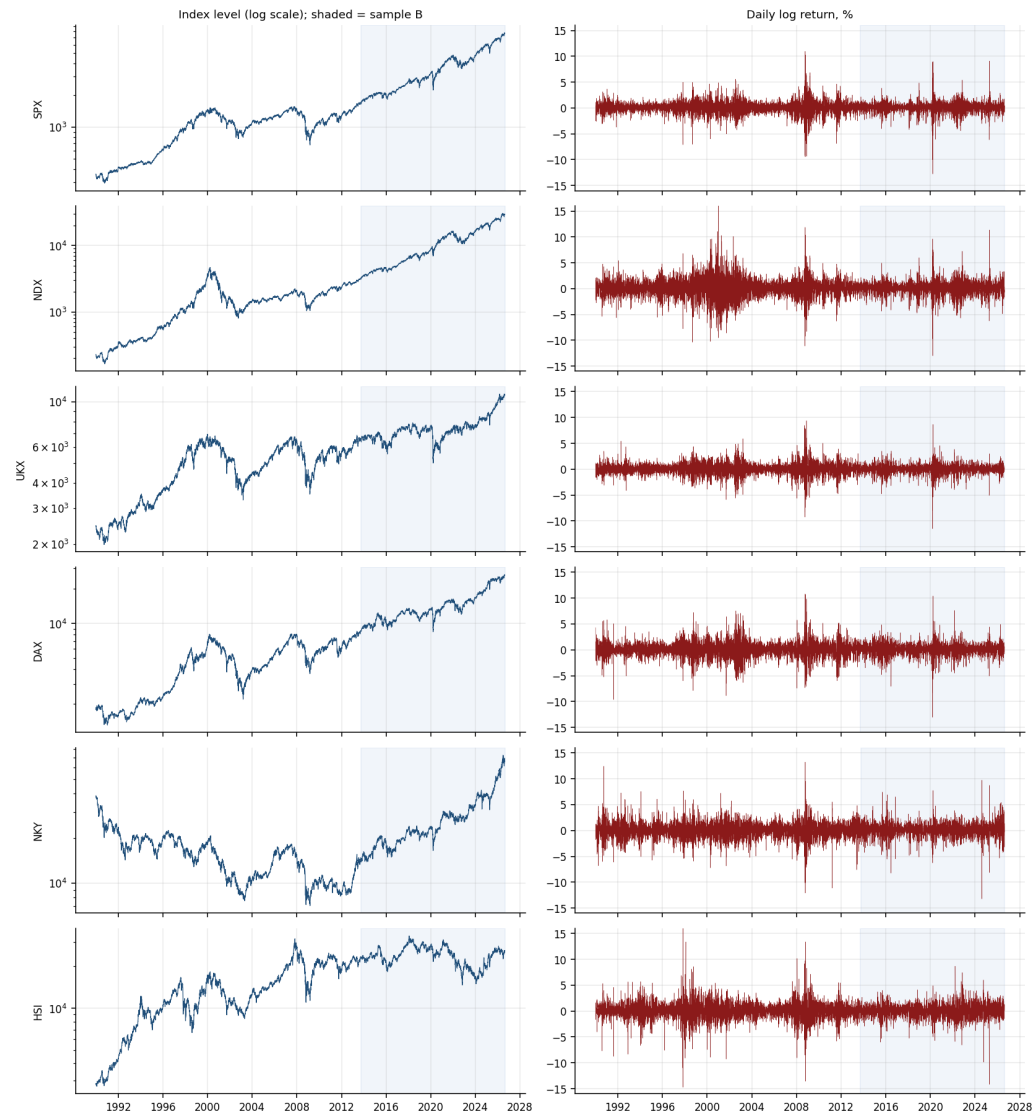


Diagnostic Figures

Exploratory data analysis — six Tier-1 equity indices

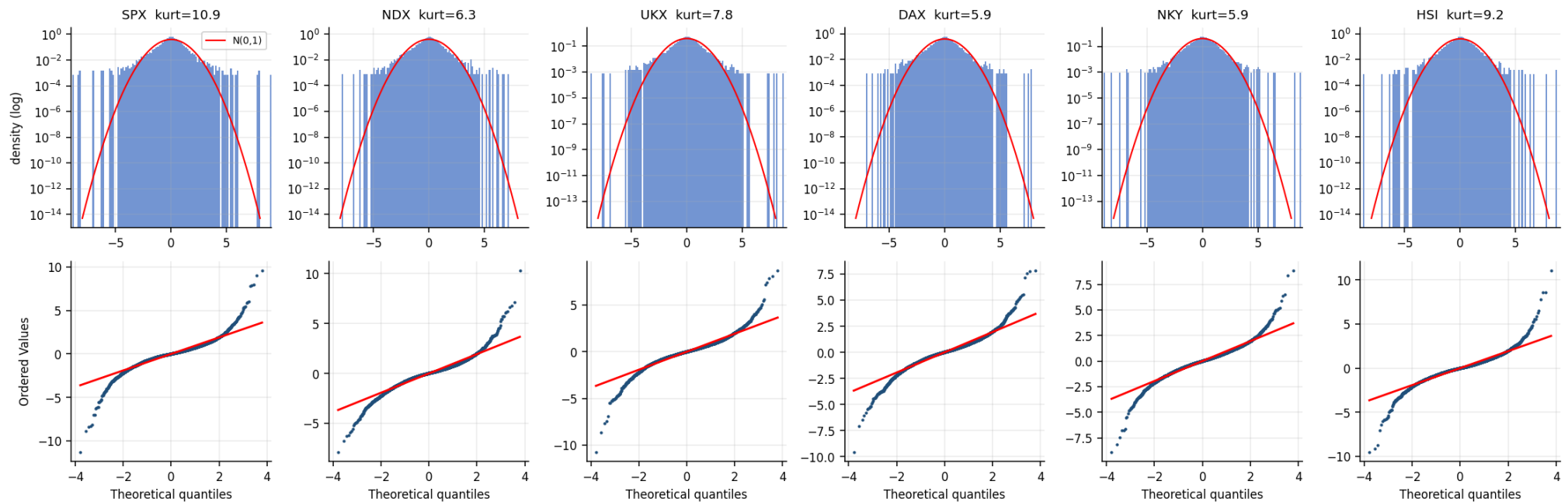
01 price return timeline



Index levels on a log scale and daily returns. The shaded band is sample B. Confirms each series is what it claims to be and locates the stress episodes.

02 return distribution qq

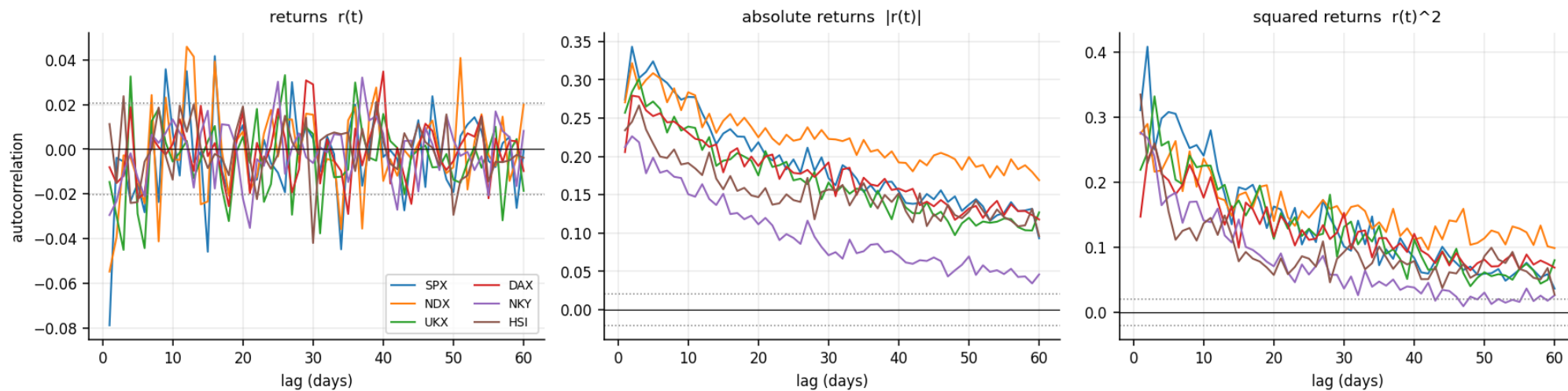
Standardised daily returns vs the normal distribution: density on a log scale (top) and normal QQ plot (bottom). Tail departure is the point.



Standardised returns against the normal. Density on a log scale (top) and normal QQ plots (bottom). The tail departure is the point — it is what a histogram alone hides and what motivates the EVT stage.

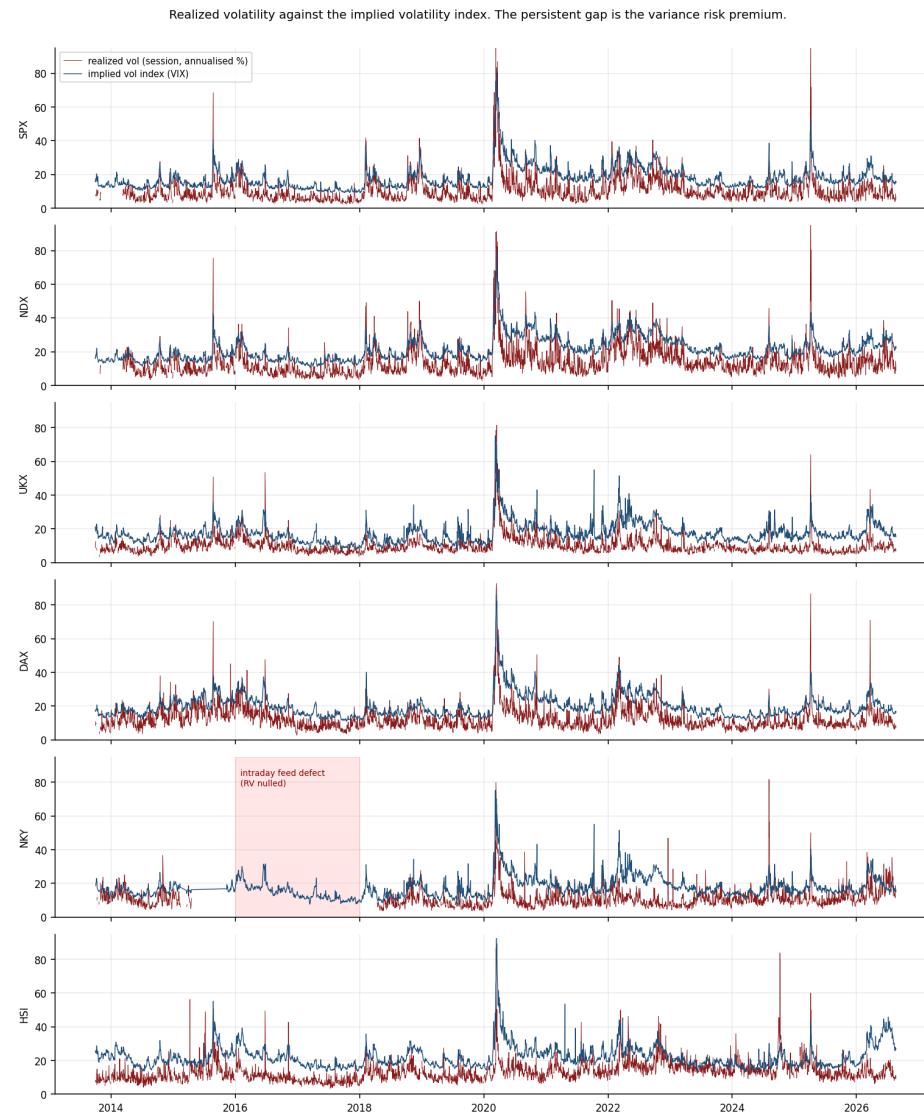
03 autocorrelation

Volatility clustering: returns are near-white, but $|r|$ and r^2 stay autocorrelated for months. This is the precondition for a GARCH model.



Volatility clustering. Returns are near-white; absolute and squared returns stay autocorrelated for months. This is the precondition for any GARCH model.

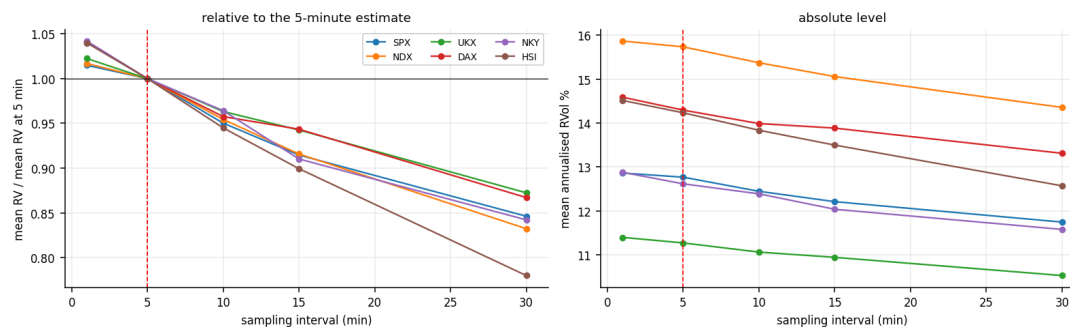
04 realized vs implied



Realized volatility against each market's implied-volatility index. The persistent gap is the variance risk premium. The red band marks the Nikkei's nulled 2016-17 feed defect.

05 volatility signature

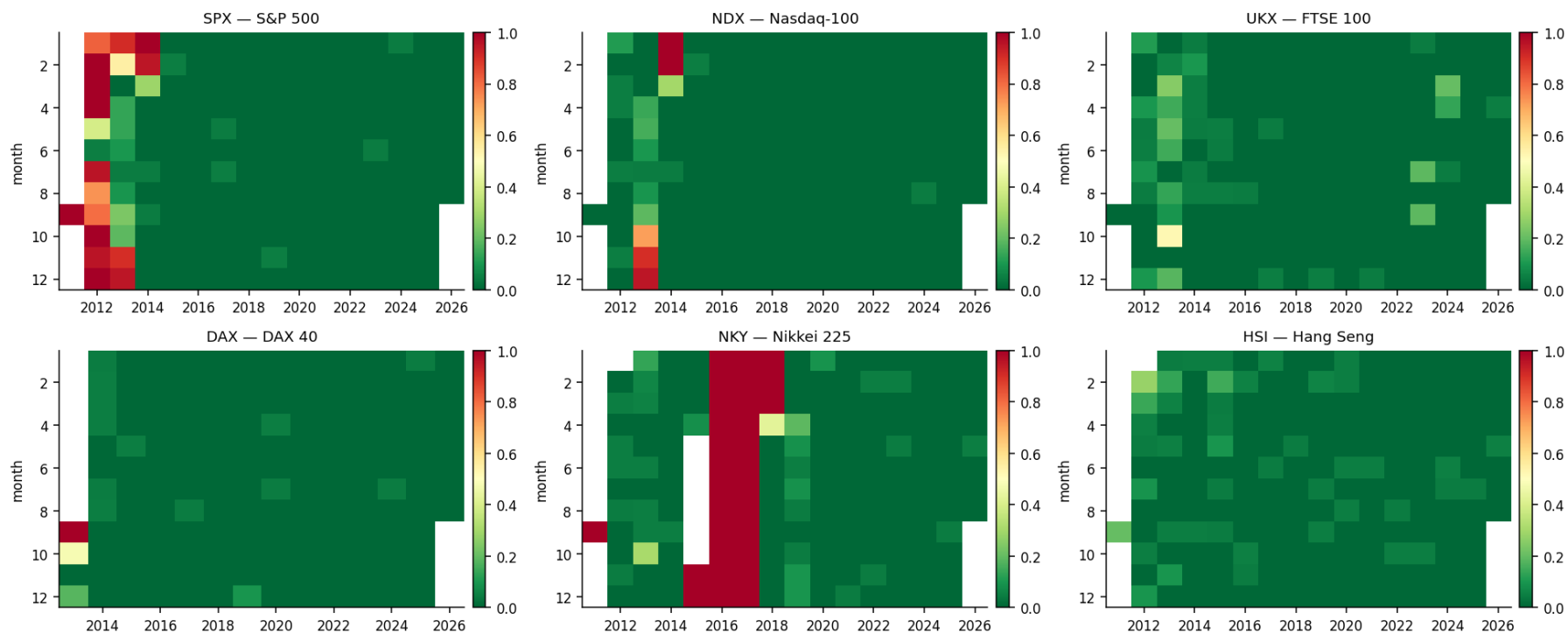
Volatility signature plot. RV declines mildly and monotonically with the sampling interval — about 15-20% in variance between 1 and 30 minutes — rather than showing the sharp high-frequency blow-up of noisy trade data. The 5-minute grid (red) sits within a few percent of the 1-minute estimate.



Volatility signature plot. RV declines mildly and monotonically with the sampling interval rather than showing the sharp high-frequency blow-up of noisy trade data. The 5-minute grid sits within a few percent of the 1-minute estimate, which is what justifies the sampling choice.

06 session quality heatmap

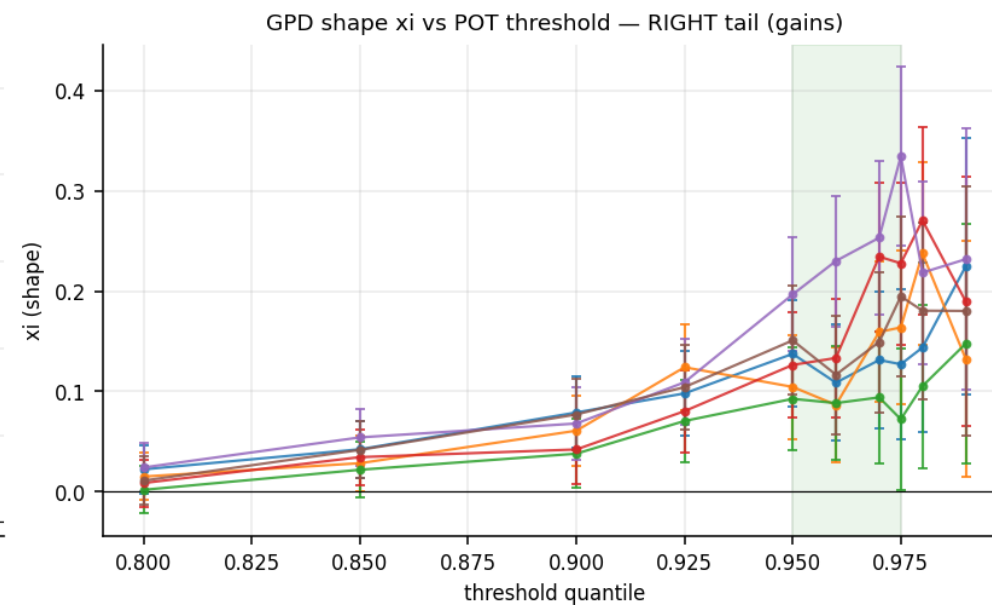
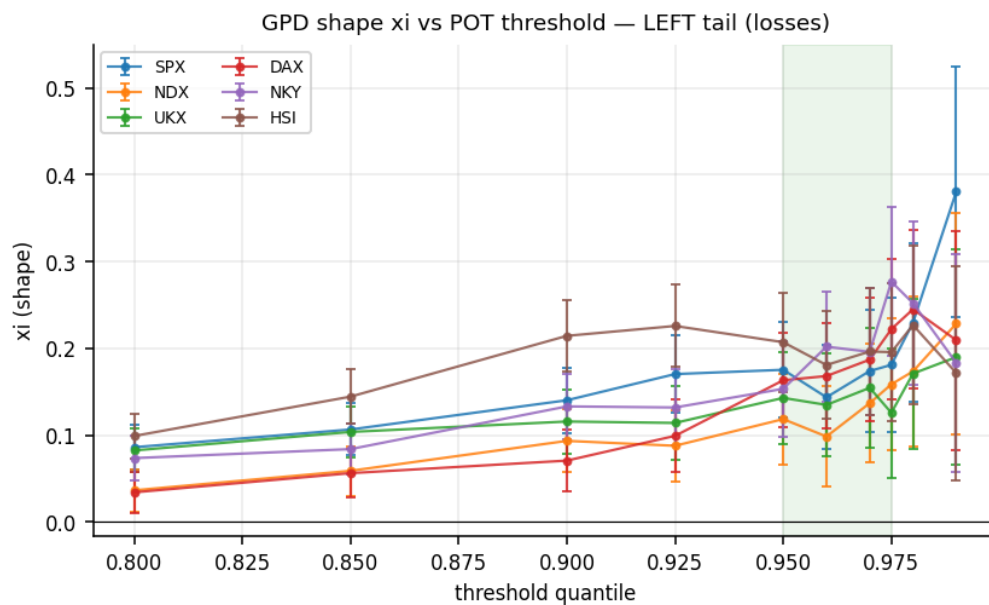
Share of each month's intraday sessions classified DEFECT (feed failure). Green = clean, red = wholly unusable. The NKY 2016-17 block is the defect that drove the cleaning rule; white = no data cached for that month.



Share of each month's intraday sessions classified DEFECT. The Nikkei 2016-17 block is the defect that drove the main cleaning rule.

07 gpd threshold stability

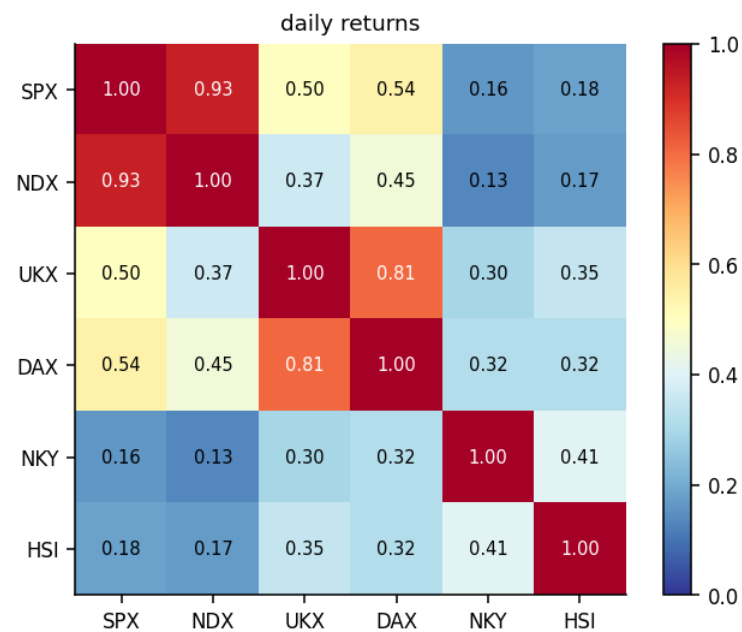
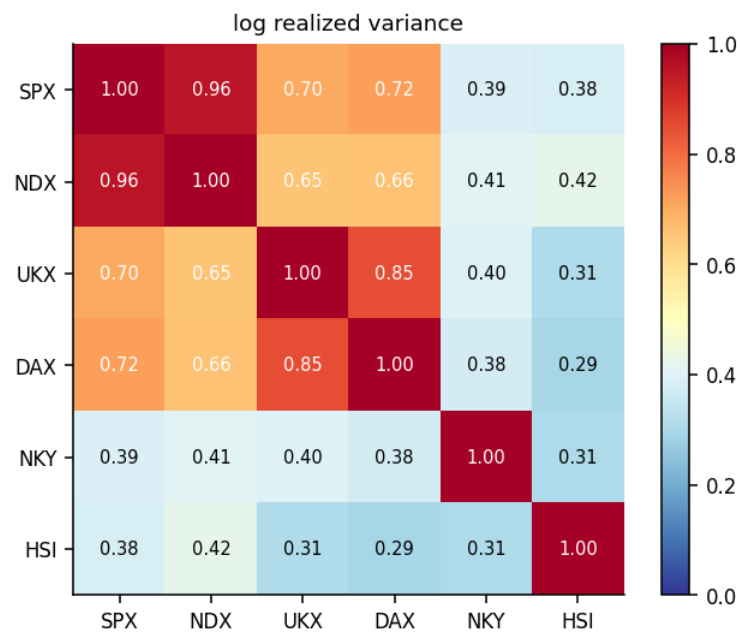
Peaks-over-threshold stability. $\xi > 0$ means a heavy (Frechet) tail even AFTER volatility standardisation. Shaded = the recommended 95-97.5% region.



Peaks-over-threshold stability. $\xi > 0$ means a heavy tail survives volatility standardisation. The shaded region is the recommended 95-97.5% threshold band.

08 cross index correlation

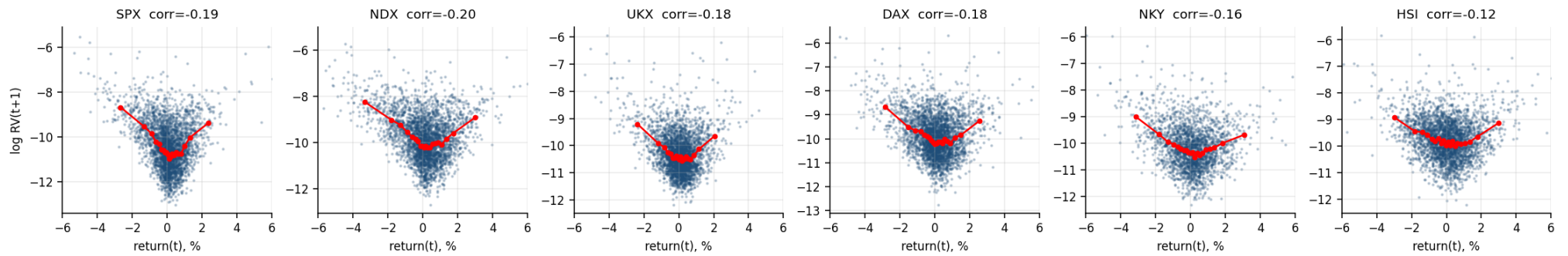
Cross-index correlation. The US and European blocks are tight; Asia is nearly uncorrelated with the US on the SAME calendar date because the sessions do not overlap.



Cross-index correlation. The US and European blocks are tight; Asia is nearly uncorrelated with the US on the same calendar date because the sessions do not overlap — a timing artefact, not independence.

09 leverage news impact

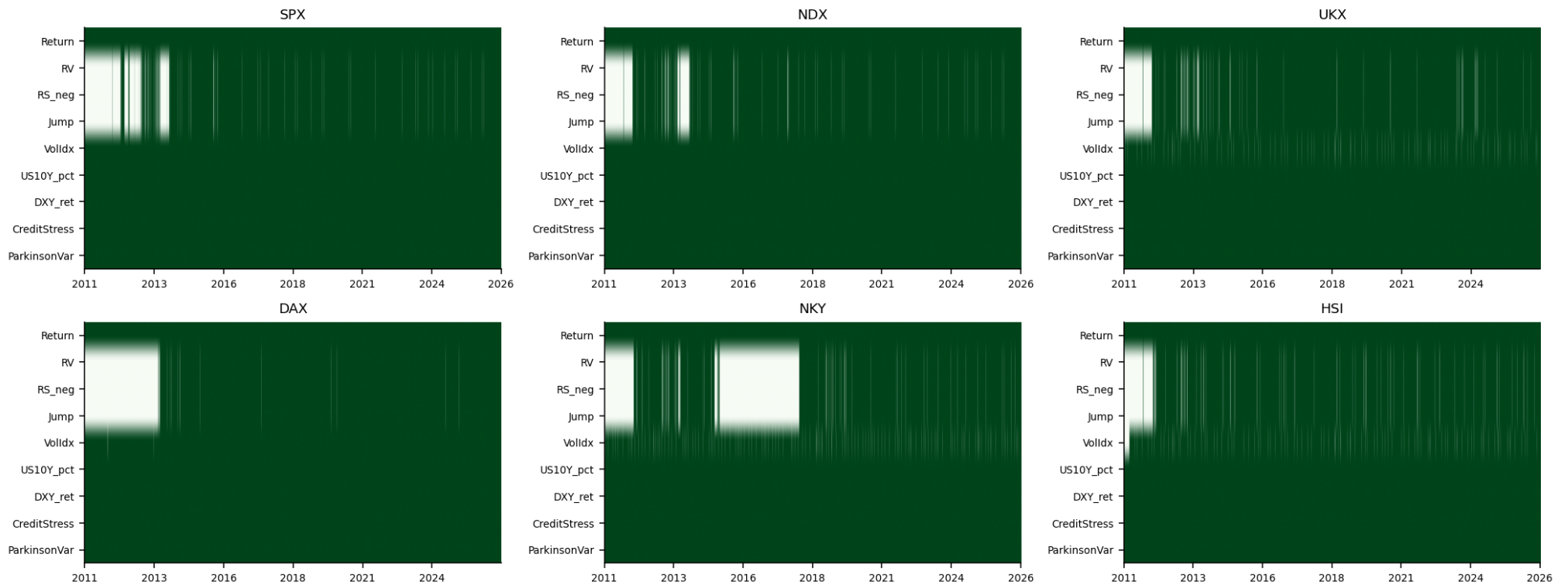
News impact: tomorrow's realized variance against today's signed return. The red binned mean is asymmetric — negative shocks raise volatility more. This is why GJR/EGARCH is required over plain GARCH.



News impact. Tomorrow's realized variance against today's signed return, with a binned mean in red. The asymmetry — negative shocks raise volatility more — is why GJR or EGARCH is required over plain GARCH.

10 missing data map

Data availability map, 2011 onward. Green = present. The RV row for NKY shows the nulled 2016-17 block; every other series is continuous.



Data availability by column over time. The RV row for the Nikkei shows the nulled 2016-17 block; every other series is continuous.